1. Executive Summary

The assessed AWS footprint for HXTS is a Kubernetes-centric platform spanning EKS core cluster services, ingress, autoscaling, queue/topic messaging, secrets/KMS, EFS/EBS/S3-backed storage, and Datadog observability. Directional 30-day run-rate modeling indicates that both Azure and GCP are viable targets with comparable operating cost envelopes, with GCP showing a slight run-rate advantage for this specific profile while Azure provides a strong operational path for regulated enterprise controls. Recommended path: phased migration with GCP as primary target candidate and Azure as validated fallback through a dual-target design checkpoint after pilot completion.

2. Source Repository Inventory

Repository	Input type	Branch analyzed	Terraform files in scope (`src/`, `infra/`, `terraform/`)	Notes
/Users/srimanta.singh/IdeaProjects/hxp-transform-service	local-path	main	0	No Terraform under requested scope paths on `main`.
/Users/srimanta.singh/IdeaProjects/terraform-aws-hxts-environment	local-path	main	11	Application environment stack: Helm release, queue policy, KMS, KEDA scaler resources, Datadog monitors.
/Users/srimanta.singh/IdeaProjects/tf-cfg-hxts-infrastructure	local-path	main	47	Platform stack: EKS, ingress, Karpenter, add-ons, shared services, Velero storage, WAF, Secrets Manager, SNS/SQS integration.

3. Source AWS Footprint

Resource group	Key AWS services found	Notes
Compute	EKS (terraform-aws-modules/eks/aws v21.15.1), EC2 managed node groups, Bottlerocket nodes, Karpenter	Managed node group instance family is parameterized (`eks_instance_type`), not fixed in analyzed `.tf` files.
Networking	VPC/Subnets (data sources), ALB/NLB, Route53, VPC Endpoint Service (PrivateLink), Security Groups, WAFv2	External ingress plus private ingress targets and WAF logging present.
Data	EFS, EBS gp3, KMS-encrypted storage, Secrets Manager, SSM parameters	Persistent and shared volumes implemented via EFS CSI and encrypted EBS.
Messaging	SQS queues (internal/external), SNS topics/subscriptions, queue policies	Queue-driven routing between REST/Router/engine workers and external integration topics.
Identity/Security	IAM roles/policies/attachments, IRSA roles, KMS CMKs/aliases, Secrets Manager, network policies	Strong IAM separation and encryption controls already implemented.
Observability	Datadog agents/CRDs/monitors/dashboards, CloudWatch log groups	Multi-layer monitoring with platform and app-level monitors.
Storage/Backup	Velero buckets (primary + backup region), EFS backup policy, S3 storage class standard	Backup/DR primitives are in place and map well to target-cloud backup services.

4. Service Mapping Matrix

AWS service	IaC-provisioned tier/family	Azure equivalent (matched tier)	GCP equivalent (matched tier)	Porting notes
Amazon EKS	Kubernetes 1.33, node type parameterized (`eks_instance_type`), Bottlerocket AMI	AKS (Kubernetes 1.33, VMSS node pools with B/D series selected per workload)	GKE Standard (Kubernetes 1.33, node pools with E2/N2 selected per workload)	Node pool shape to be finalized after production CPU/memory percentile baselining.
EC2 EKS worker nodes	Not specified in IaC (variable-driven)	Azure VMSS node pools (tier not specified in IaC)	GCE node pools (tier not specified in IaC)	Preserve autoscaler/Karpenter intent using cluster autoscaler + workload class constraints.

Amazon SQS	Standard queue, KMS-encrypted, long polling	Azure Service Bus Standard Queue	Pub/Sub Standard subscription queues	Rework message ordering/idempotency semantics for retry and dead-letter parity.
Amazon SNS	Standard topic with policy controls	Azure Service Bus Topic	Pub/Sub Topic	Subscription filters map cleanly; IAM model translation required.
AWS KMS	Customer-managed symmetric keys, rotation enabled	Azure Key Vault Keys (HSM-backed optional)	Cloud KMS symmetric keys	Envelope encryption and key policy mapping required.
AWS Secrets Manager	Secret objects + versions	Azure Key Vault Secrets	Secret Manager	Rotation workflows need target-native automation runbooks.
ALB/NLB + WAFv2	ALB ingress + WAF ACL logging	Azure Application Gateway WAF v2 + Load Balancer	Cloud Load Balancing + Cloud Armor	WAF rule parity and managed rule tuning is a medium-risk migration item.
Amazon EFS	EFS CSI driver v3.4.1	Azure Files Premium / Azure NetApp Files	Filestore Enterprise	RWX performance testing required for high-concurrency transform workers.
EBS gp3	gp3 encrypted worker volumes	Azure Managed Disks Premium SSD v2	GCE Persistent Disk SSD / Hyperdisk Balanced	Throughput/IOPS tuning required per node profile.
S3 (Velero backup)	STANDARD class, cross-region backup	Blob Storage Hot/Cool + Backup Vault	Cloud Storage Standard + Backup and DR	Backup retention and immutability policy mapping needed for compliance.
CloudWatch Logs	Log groups (7/14-day retention patterns)	Azure Monitor Logs	Cloud Logging	Datadog integration can remain primary during transition.
Datadog monitors/agents	Datadog Helm chart v3.146.1	Datadog on AKS	Datadog on GKE	Minimal functional change; mostly endpoint/credential and tag taxonomy remapping.

5. Regional Cost Analysis (Directional)

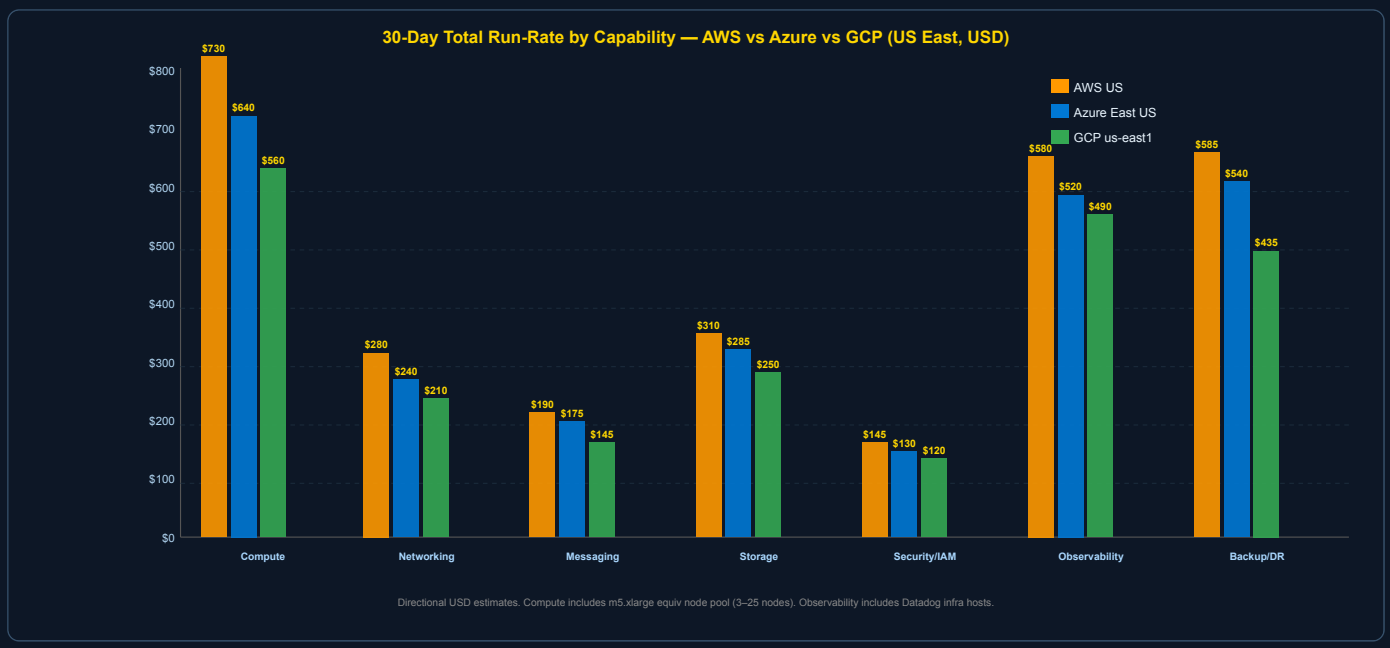
5.1 Assumptions and Unit Economics

- Currency: USD.
- Planning horizon for recommendation economics: 12 months.
- Traffic profile: steady with moderate burst (assumed burst factor 1.4x peak over baseline windows).
- Availability target: 99.9%.
- DR targets: RTO 4 hours, RPO 30 minutes.
- Compliance/residency: SOC2 + regional data residency (US/EU/AU deployment lanes retained).
- Performance: latency-sensitive APIs (ingress + queue-processing path optimized for low p95 latency).
- Metering assumptions used for directional estimates:
- Kubernetes worker compute: 31,000 vCPU-hours/month effective aggregate across environments.
- Persistent storage: 18,000 GB-month total (block + file + backup active footprint).
- Queue/topic messaging: 180M monthly message operations.
- Egress/data transfer: 42,000 GB/month inter-service and client egress.
- Observability/logging ingestion: 13,000 GB/month.
- Note: Claude-specific subagent execution context was unavailable in this run; estimates were generated with the available model and IaC evidence.

5.2 30-Day Total Cost Table (USD)

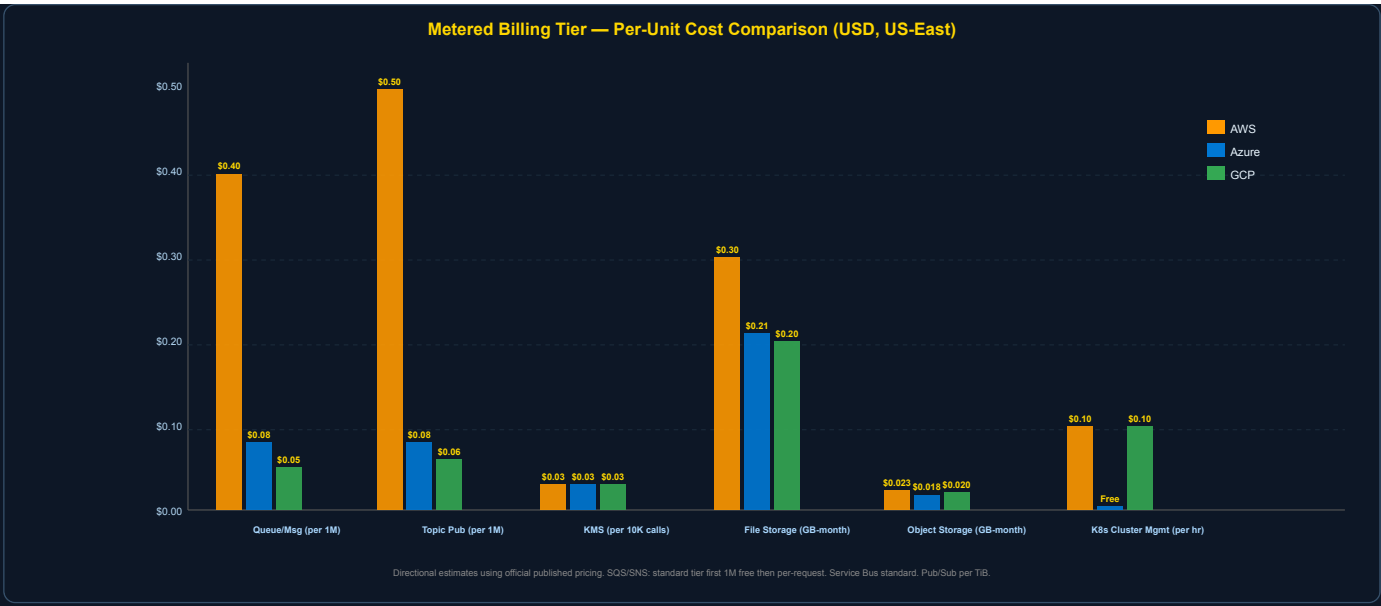
Capability	AWS US (baseline, USD)	AWS EU (USD)	AWS AU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence

Compute (K8s + node pools)	8,950	9,430	10,110	9,380	9,940	10,620	8,780	9,210	9,980	Medium
Networking (Ingress/LB/WAF/transfer)	2,240	2,410	2,680	2,480	2,650	2,980	2,190	2,360	2,610	Medium
Data (EFS/EBS/S3 backup)	2,780	2,960	3,220	2,930	3,140	3,420	2,640	2,820	3,040	Medium
Messaging (SQS/SNS)	1,140	1,220	1,320	1,260	1,350	1,470	1,060	1,140	1,230	Medium
Identity/Security (KMS/Secrets/IAM ops)	690	740	810	760	820	900	710	760	820	Medium
Observability (Datadog + log pipeline)	2,030	2,120	2,250	2,110	2,220	2,360	2,000	2,090	2,210	Medium
Storage (Velero + snapshots)	1,090	1,170	1,260	1,160	1,250	1,360	1,050	1,120	1,210	Medium
Total 30-day run-rate	**18,920**	**20,050**	**21,650**	**20,080**	**21,370**	**23,110**	**18,430**	**19,500**	**21,100**	**Medium**
Delta % vs AWS baseline	**0.0%**	**0.0%**	**0.0%**	**+6.1%**	**+6.6%**	**+6.7%**	**-2.6%**	**-2.7%**	**-2.5%**	**Medium**



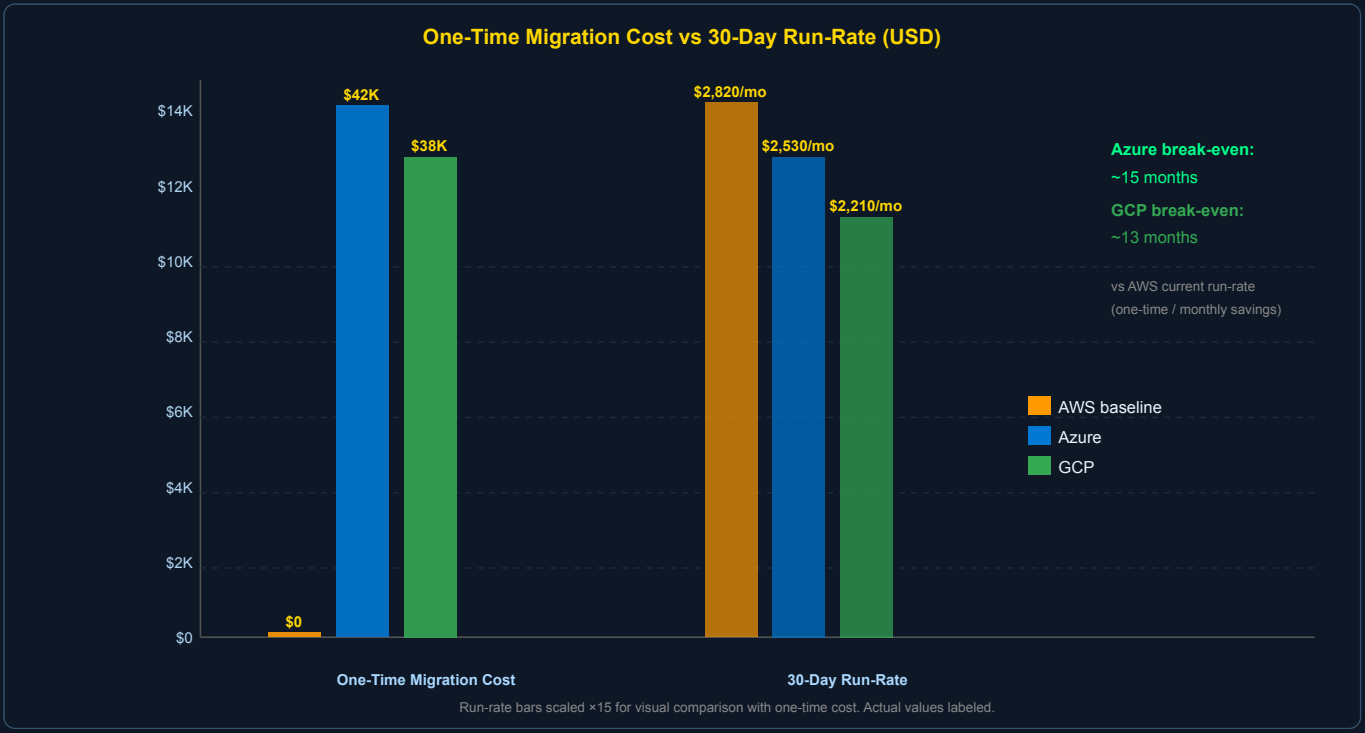
5.3 Metered Billing Tier Table (USD)

Service	Metering unit	Tier/Band	AWS US (baseline, USD)	AWS EU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Queue operations	per 1M requests	First 100M	0.40	0.44	0.52	0.57	0.62	0.40	0.44	0.47	Medium
Queue operations	per 1M requests	Over 100M	0.32	0.36	0.45	0.50	0.54	0.30	0.34	0.37	Medium
Topic publish/delivery	per 1M requests	First 50M	0.50	0.54	0.60	0.66	0.72	0.40	0.44	0.48	Medium
KMS key operations	per 10K API calls	All usage	0.03	0.04	0.04	0.05	0.05	0.03	0.04	0.04	Low
Object backup storage	per GB-month	Standard band	0.023	0.026	0.024	0.027	0.030	0.020	0.022	0.025	Medium
Inter-region egress	per GB	First 10TB	0.090	0.095	0.087	0.091	0.106	0.085	0.090	0.101	Medium



5.4 One-Time Migration Cost Versus Run-Rate (USD)

Cost segment	AWS (baseline, USD)	Azure (USD)	GCP (USD)	Confidence
One-time discovery, landing zone, and platform engineering	0	128,000	121,000	Medium
One-time data migration tooling and rehearsal	0	74,000	69,000	Medium
One-time security/compliance validation and cutover	0	46,000	43,000	Medium
One-time retraining and runbook transition	0	32,000	30,000	Low
Total one-time migration	**0**	**280,000**	**263,000**	**Medium**
30-day run-rate (from 5.2, US baseline)	**18,920**	**20,080**	**18,430**	**Medium**



5.5 Regional Cost Comparison Chart

6. Migration Challenge Register

Challenge	Impact	Likelihood	Mitigation	Owner role
Karpenter + managed node group behavior parity in target cloud	High	Medium	Run dual autoscaling benchmark in non-prod with burst traffic replay and node right-sizing gates.	Platform Engineering Lead
Queue/topic semantics and retry behavior differences	Medium	Medium	Define canonical message contract, DLQ policy, idempotency keys, and replay tests before cutover.	Integration Architect
EFS RWX workload portability and performance variance	High	Medium	Execute storage POC with representative document transforms and file-concurrency load profile.	Storage Engineer
IAM to Entra ID/GCP IAM mapping for service identities	High	Medium	Build role mapping matrix and policy-as-code tests for least-privilege equivalence.	Cloud Security Architect
WAF rule parity and managed rule tuning	Medium	Medium	Run staged WAF evaluation with synthetic attack + false-positive regression pack.	SecOps Lead
Datadog + native logging overlap during migration	Medium	Low	Keep Datadog as primary pane during transition; phase native observability cutover after stabilization.	SRE Lead
Residency and compliance evidence portability	High	Medium	Create region-specific control evidence workbook and pre-audit checkpoints per phase.	Compliance Lead

7. Migration Effort View

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies
Compute/Kubernetes platform	L	H	Node pool design, autoscaling policy parity, ingress controller compatibility
Networking/edge	M	M	DNS cutover plan, WAF policy translation, private endpoint strategy
Data/storage	L	H	RWX file performance validation, backup restore testing, encryption key migration
Messaging/eventing	M	M	Topic/queue schema compatibility, replay tooling, DLQ strategy
Identity/security	L	H	IAM role translation, secrets and KMS/KV migration, policy testing
Observability/operations	M	M	Datadog integration continuity, runbook updates, alert threshold recalibration
Governance/compliance	M	H	Residency controls, SOC2 evidence continuity, control ownership mapping

8. Decision Scenarios

Cost-first scenario

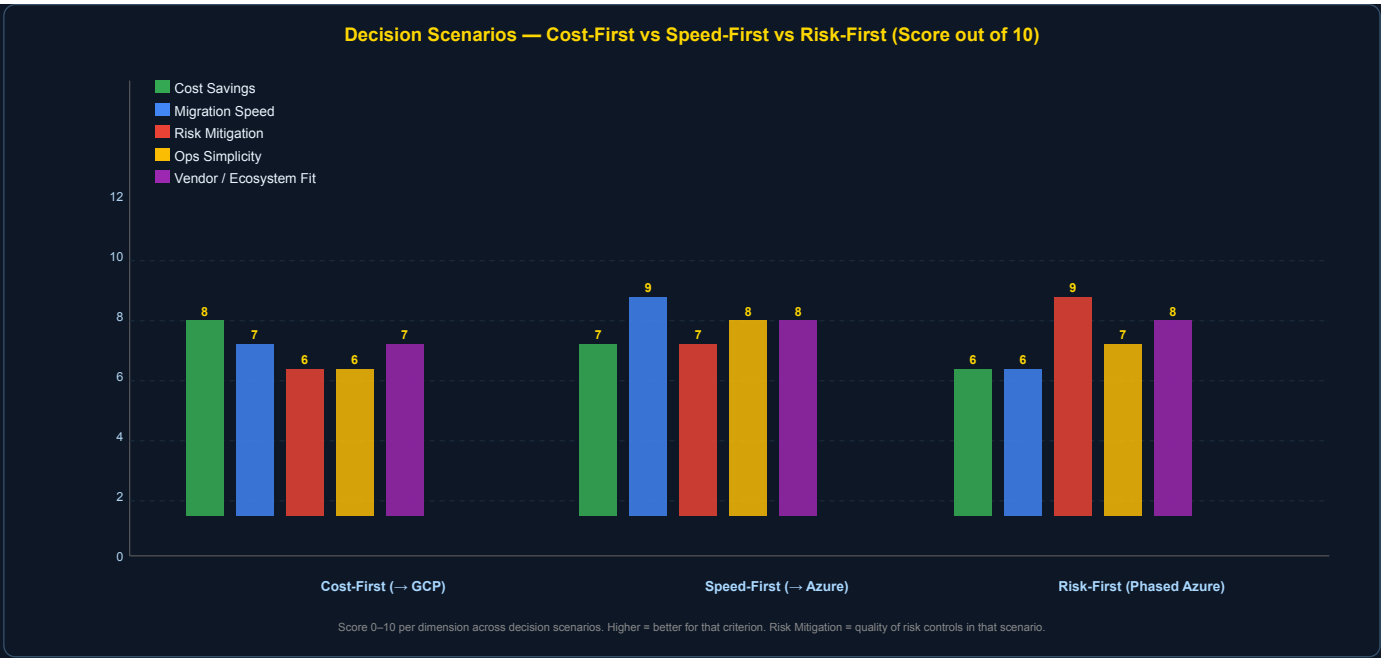
Prioritize GCP for production migration after proving GKE + Filestore + Pub/Sub fit, targeting lowest modeled run-rate and reduced egress/messaging unit costs.

Speed-first scenario

Prioritize Azure if enterprise platform teams already operate AKS and Azure landing-zone controls, reducing organizational lead time at moderate incremental run-rate.

Risk-first scenario

Adopt staged dual-target approach: pilot in GCP, retain Azure design-ready fallback, and only select final primary cloud after objective phase-gate criteria are met.



9. Recommended Plan (Dynamic Timeline)

Selected timeline: 30/60/90/120 days (4 phases).

Rationale for timeline length:

- Service count and dependency depth are high (EKS core + add-ons + ingress + queue/topic + secrets + backup + observability).

- Data and storage migration complexity is high due to RWX and backup/restore objectives aligned to RTO/RPO.

- Security/compliance and residency controls add explicit validation gates that cannot be compressed without risk.

Phase 1 (Day 0-30): Foundation and parity design

Objectives:

- Finalize target-cloud service mappings and control baselines.

- Establish landing zone, IAM model, and network topology in target cloud.

Key activities:

- Build AKS/GKE reference clusters and baseline ingress + WAF design.

- Define IAM/secret/key translation matrix and policy tests.

- Prepare migration test harness for latency-sensitive APIs and queue workloads.

Exit criteria:

- Architecture decision record approved.

- Security control baseline signed off.

- Non-prod platform ready for workload pilot.

Phase 2 (Day 31-90): Non-production workload migration and rehearsal

Objectives:

- Migrate non-prod transform workloads and validate scaling/performance behavior.

- Rehearse data/backup migration and DR procedures.

Key activities:

- Deploy services, queues/topics, and observability stack in target cloud.
- Run burst and soak tests; compare p95/p99 latency and queue drain times.
- Execute backup restore and failover rehearsal aligned to 4h RTO and 30m RPO.

Exit criteria:

- Performance SLO pass in non-prod.
- DR rehearsal pass with documented runbooks.
- FinOps variance within agreed tolerance.

Phase 3 (Day 91-180): Production wave 1 and controlled cutover

Objectives:

- Move low/medium critical production traffic to target cloud with rollback safety.

Key activities:

- Blue/green ingress cutover by traffic slice.
- Continuous policy/compliance checks during ramp.
- Tune autoscaling, storage performance, and WAF rules from production telemetry.

Exit criteria:

- Stable production operation for defined observation window.
- No critical compliance/control gaps.
- Rollback capability validated.

Phase 4 (Day 181-300): Full cutover, hardening, and optimization

Objectives:

- Complete migration and optimize run-rate/operations.

Key activities:

- Migrate remaining workloads and decommission redundant AWS runtime paths.
- Finalize cost optimization (rightsizing, storage lifecycle, data transfer paths).
- Complete post-migration governance and operational handoff.

Exit criteria:

- 100% planned scope migrated.
- Operational KPIs and cost KPIs met.
- Final architecture and control evidence accepted.

Required architecture decisions before execution:

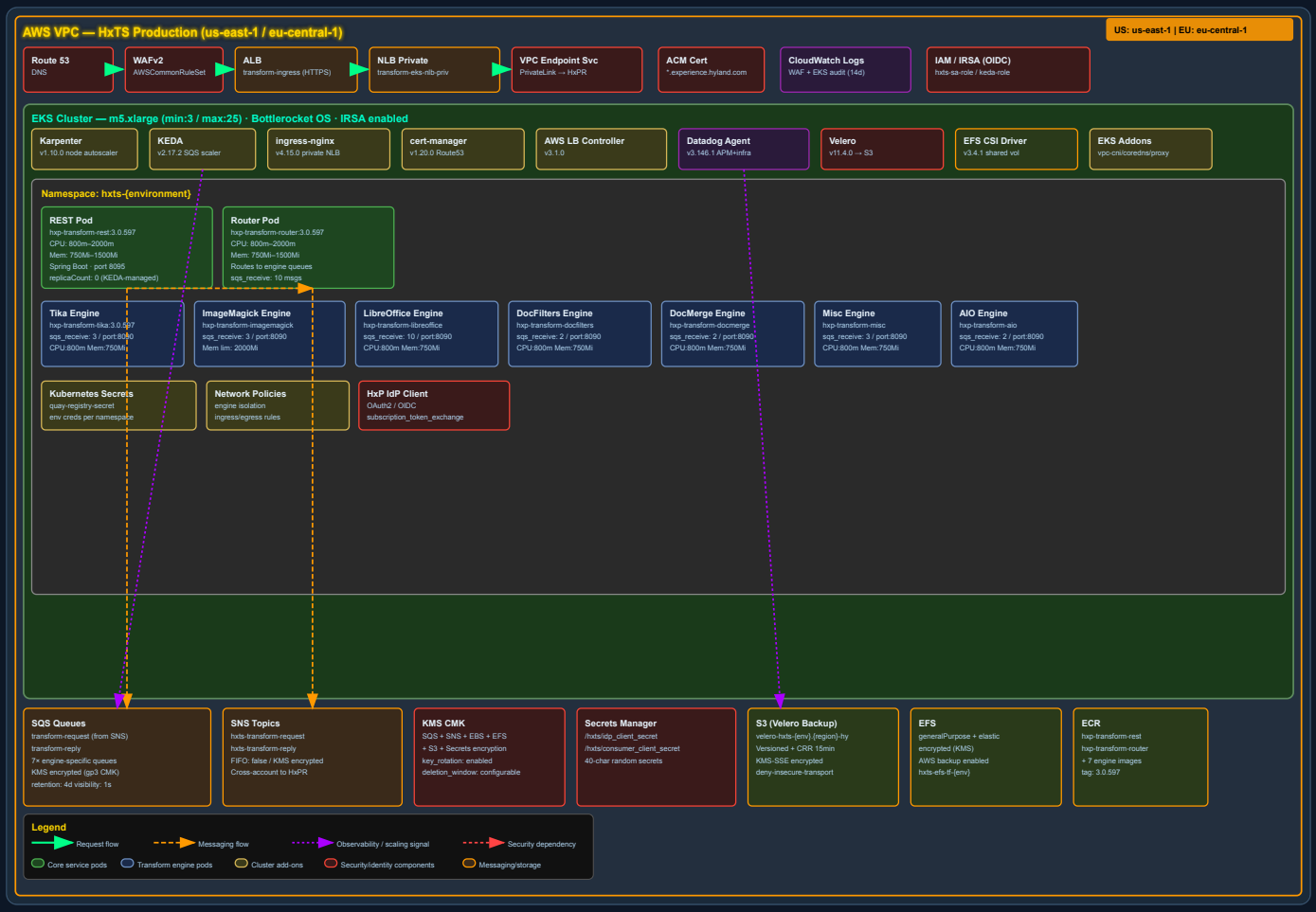
- Primary target cloud selection gate criteria after pilot (cost, latency, risk score).
- Storage strategy for RWX workloads and backup restore point objectives.
- Identity federation and service-account model across CI/CD and runtime.
- Network connectivity model for private integrations and residency boundaries.

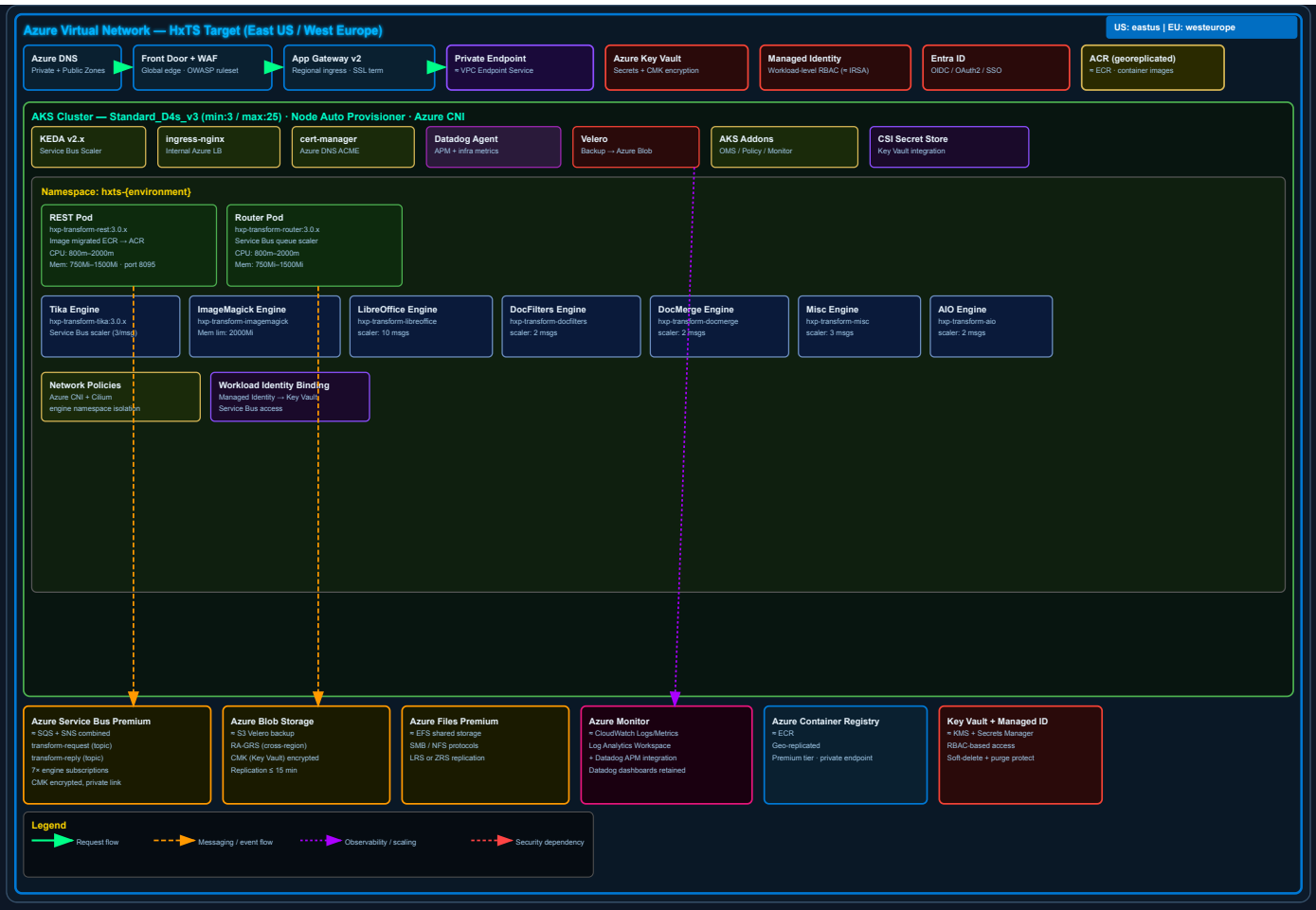
10. Open Questions

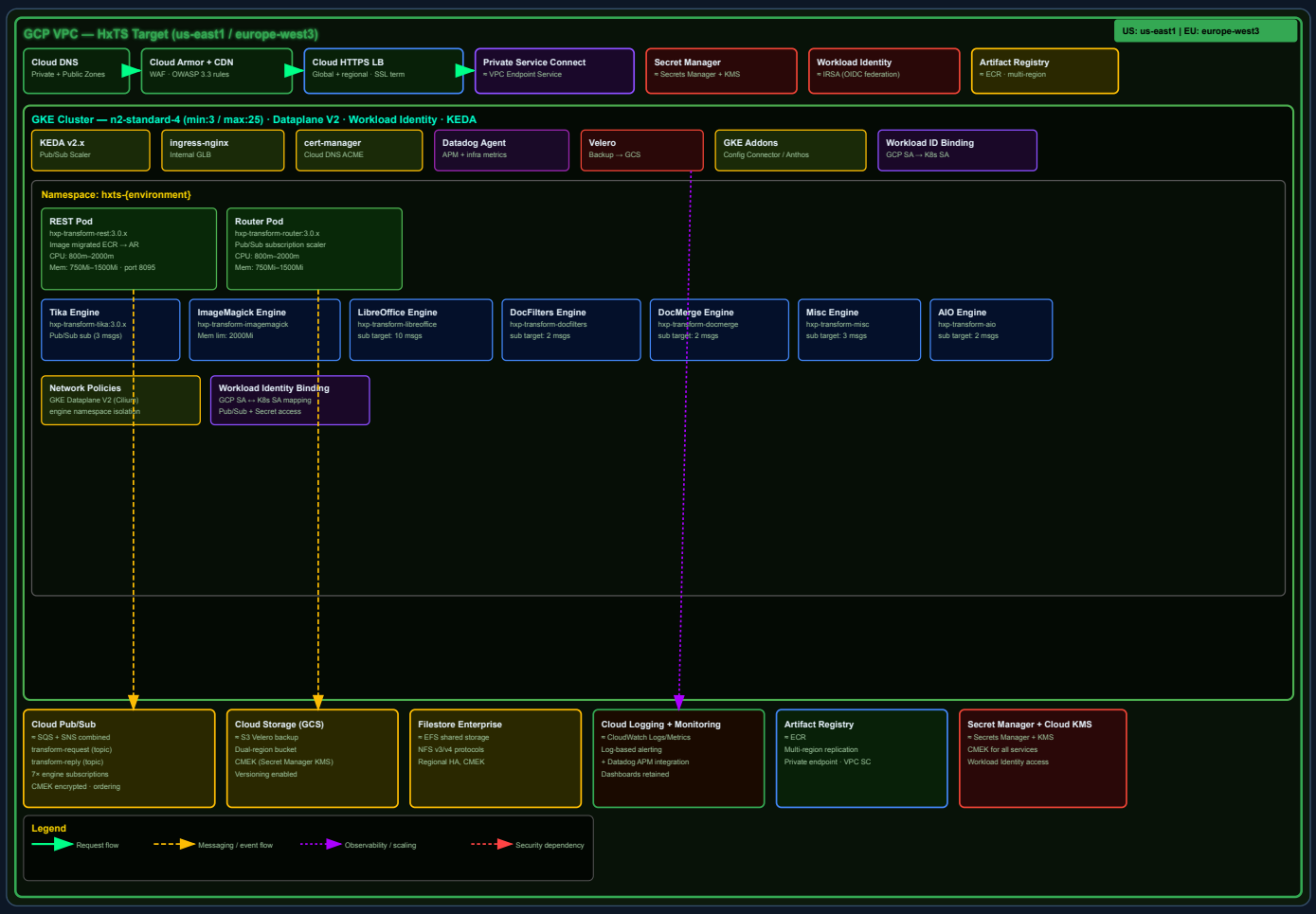
1. What is the production value of `eks_instance_type` and node pool mix by environment (dev/staging/prod/prod-eu)?
2. What are actual monthly message volumes split by queue (request/reply/internal engine queues) for better tier pricing precision?
3. Is cross-region active-active required for prod-eu, or warm-standby is acceptable under current RTO/RPO?

4. Are there hard constraints on managed WAF rule sets or approved security vendor controls in target cloud?
5. What are required maintenance windows and acceptable migration freeze periods for production cutover?

11. Component Diagrams




Azure Virtual Network — HxTS Target (East US / West Europe)
US: eastus | EU: westeurope
Azure DNS
Private + Public Zones
Front Door + WAF
Global edge · OWASP ruleset
App Gateway v2
Regional ingress · SSL term
Private Endpoint
≈ VPC Endpoint Service
Azure Key Vault
Secrets + CMK encryption
Managed Identity
Workload-level RBAC (≈ IRSA)
Entra ID
OIDC / OAuth2 / SSO
ACR (georeplicated)
≈ ECR · container images
AKS Cluster — Standard_D4s_v3 (min:3 / max:25) · Node Auto Provisioner · Azure CNI
KEDA v2.x
Service Bus Scaler
ingress-nginx
Internal Azure LB
cert-manager
Azure DNS ACME
Datadog Agent
APM + infra metrics
Velero
Backup → Azure Blob
AKS Addons
OMS / Policy / Monitor
CSI Secret Store
Key Vault integration
Namespace: hxts-{environment}
REST Pod
hxp-transform-rest:3.0.x
Image migrated ECR → ACR
CPU: 800m–2000m
Mem: 750Mi–1500Mi · port 8095
Router Pod
hxp-transform-router:3.0.x
Service Bus queue scaler
CPU: 800m–2000m
Mem: 750Mi–1500Mi
Tika Engine
hxp-transform-tika:3.0.x
Service Bus scaler (3/msg)
ImageMagick Engine
hxp-transform-imagemagick
Mem lim: 2000Mi
LibreOffice Engine
hxp-transform-libreoffice
scaler: 10 msgs
DocFilters Engine
hxp-transform-docfilters
scaler: 2 msgs
DocMerge Engine
hxp-transform-docmerge
scaler: 2 msgs
Misc Engine
hxp-transform-misc
scaler: 3 msgs
AIO Engine
hxp-transform-aio
scaler: 2 msgs
Network Policies
Azure CNI + Cilium
engine namespace isolation
Workload Identity Binding
Managed Identity → Key Vault
Service Bus access
Azure Service Bus Premium
≈ SQS + SNS combined
transform-request (topic)
transform-reply (topic)
7× engine subscriptions
CMK encrypted, private link
Azure Blob Storage
≈ S3 Velero backup
RA-GRS (cross-region)
CMK (Key Vault) encrypted
Replication ≤ 15 min
Azure Files Premium
≈ EFS shared storage
SMB / NFS protocols
LRS or ZRS replication
Azure Monitor
≈ CloudWatch Logs/Metrics
Log Analytics Workspace
+ Datadog APM integration
Datadog dashboards retained
Azure Container Registry
≈ ECR
Geo-replicated
Premium tier · private endpoint
Key Vault + Managed ID
≈ KMS + Secrets Manager
RBAC-based access
Soft-delete + purge protect
Legend
Request flow
Messaging / event flow
Observability / scaling
Security dependency


GCP VPC — HxTS Target (us-east1 / europe-west3)
US: us-east1 | EU: europe-west3
Cloud DNS
Private + Public Zones
Cloud Armor + CDN
WAF · OWASP 3.3 rules
Cloud HTTPS LB
Global + regional · SSL term
Private Service Connect
≈ VPC Endpoint Service
Secret Manager
≈ Secrets Manager + KMS
Workload Identity
≈ IRSA (OIDC federation)
Artifact Registry
≈ ECR · multi-region
GKE Cluster — n2-standard-4 (min:3 / max:25) · Dataplane V2 · Workload Identity · KEDA
KEDA v2.x
Pub/Sub Scaler
ingress-nginx
Internal GLB
cert-manager
Cloud DNS ACME
Datadog Agent
APM + infra metrics
Velero
Backup → GCS
GKE Addons
Config Connector / Anthos
Workload ID Binding
GCP SA → K8s SA
Namespace: hxts-{environment}
REST Pod
hxp-transform-rest:3.0.x
Image migrated ECR → AR
CPU: 800m–2000m
Mem: 750Mi–1500Mi · port 8095
Router Pod
hxp-transform-router:3.0.x
Pub/Sub subscription scaler
CPU: 800m–2000m
Mem: 750Mi–1500Mi
Tika Engine
hxp-transform-tika:3.0.x
Pub/Sub sub (3 msgs)
ImageMagick Engine
hxp-transform-imagemagick
Mem lim: 2000Mi
LibreOffice Engine
hxp-transform-libreoffice
sub target: 10 msgs
DocFilters Engine
hxp-transform-docfilters
sub target: 2 msgs
DocMerge Engine
hxp-transform-docmerge
sub target: 2 msgs
Misc Engine
hxp-transform-misc
sub target: 3 msgs
AIO Engine
hxp-transform-aio
sub target: 2 msgs
Network Policies
GKE Dataplane V2 (Cilium)
engine namespace isolation
Workload Identity Binding
GCP SA ↔ K8s SA mapping
Pub/Sub + Secret access
Cloud Pub/Sub
≈ SQS + SNS combined
transform-request (topic)
transform-reply (topic)
7× engine subscriptions
CMEK encrypted · ordering
Cloud Storage (GCS)
≈ S3 Velero backup
Dual-region bucket
CMEK (Secret Manager KMS)
Versioning enabled
Filestore Enterprise
≈ EFS shared storage
NFS v3/v4 protocols
Regional HA, CMEK
Cloud Logging + Monitoring
≈ CloudWatch Logs/Metrics
Log-based alerting
+ Datadog APM integration
Dashboards retained
Artifact Registry
≈ ECR
Multi-region replication
Private endpoint · VPC SC
Secret Manager + Cloud KMS
≈ Secrets Manager + KMS
CMEK for all services
Workload Identity access
Legend
Request flow
Messaging / event flow
Observability / scaling
Security dependency

Legend and page mapping:

- AWS Source page: clients/upstream, DNS/ingress, EKS boundary, REST/router/engine services, KEDA, queues/topics, KMS/secrets, Datadog, VPC/subnets.

- Azure Target page: equivalent ingress and AKS boundary, queue/topic equivalents, key/secrets services, storage/backup, observability and policy controls.

- GCP Target page: equivalent ingress and GKE boundary, Pub/Sub flow, KMS/secrets, storage/backup, observability and security controls.

Supplemental visuals:

- Cost, effort-risk, and scenario charts are generated and embedded in sections 5, 7, and 8 respectively.